Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.8 Solving Problems with 3D Solids and Measures

Solving Problems with 3D Solids and Measures

STARTER · 5 MIN

Date: _____ **Title: Solving Problems with 3D Solids and Measures**

LEARNING OBJECTIVES

- Solve problems involving area, surface area and volume

STARTER *Answer in your book. No calculators.*

1. Work out 50 × 25 × 2
2. Convert 160 ml to litres
3. Work out 3 × 3 × 3
4. Work out 3.6 × 2.5

Starter — Answers

STARTER

Tick or correct in green pen

1. **2500**
2. **0.16 litres**
3. **27**
4. **9**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use the correct formula for area, surface area or volume	☐
SILVER	I can interpret worded problems	☐
GOLD	I can convert units accurately	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Solving Problems with 3D Solids and Measures

INTRODUCTION · 3 MIN

Read the question, choose the formula, then check the units.

Area

Flat space: square units.

cm^2, m^2

Surface area

Add the areas of all the faces.

$2(lw + lh + wh)$

Volume

Space inside: cubic units.

$V = lwh$

Units

Convert before you calculate.

$1\ m^3 = 1000$ litres

Examples 1 and 2 — Capacity Problems

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Same units first 1 m^3 = 1000 litres

Example 1

Ali needs 160 ml of medicine.
A bottle holds 0.15 litres.

Is the bottle big enough?

Example 2

A pool is 50 m by 25 m
by 2 m.

How many litres does it hold?

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Use the same units

0.15 litres $= 0.15 \times 1000$

$= 150$ ml

No: 150 ml < 160 ml

Tip: change litres to ml.

Example 2

Volume, then litres

$V = 50 \times 25 \times 2$

$= 2500$ m^3

= 2 500 000 litres

Tip: 1 m^3 = 1000 litres.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve*

Hint Same units first 1 m^3 = 1000 litres

1. Is 0.3 litres more than 280 ml?
2. Jug 1.5 litres: how many 250 ml cups?
3. Tank 2 m by 1 m by 1 m. Litres?
4. Bottle 0.75 litres: how many 150 ml glasses?
5. Pool 10 m by 5 m by 1.5 m. Litres?

★ 2 litres from 330 ml cans: how many cans?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **Yes: 300 ml**
2. **6 cups**
3. **2000 litres**
4. **5 glasses**
5. **75 000 litres**

★ **6.06…, so 7 cans**

Examples 3 and 4 — Compound Solids and Cubes

EXAMPLES · 5 MIN

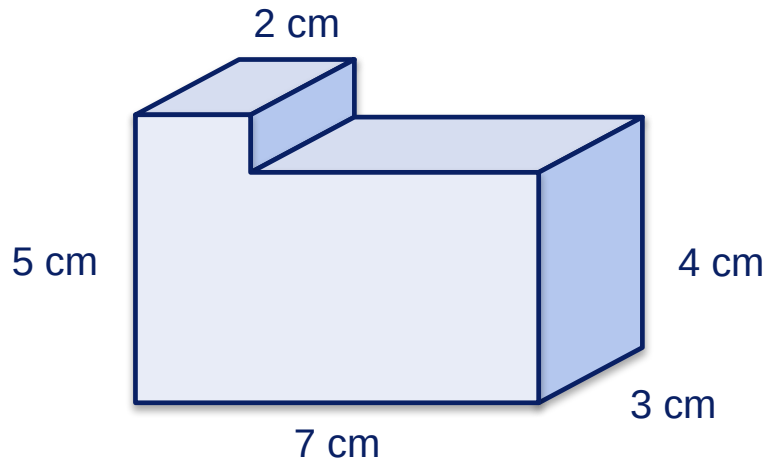
SILVER

Watch and copy into your book

Hint Cube: find the side first Prism: area × length

Example 3

Find the volume of this prism.



Example 4

A cube has volume 27 cm³.

Find its surface area.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Area of the L-shape × length

area $= 2 \times 5 + 5 \times 4 = 30\ \text{cm}^2$

$V = 30 \times 3$

$\mathbf{= 90\ cm^3}$

Tip: or add two cuboids: 30 + 60.

Example 4

Find the side first

side $= 3\ \text{cm}$

SA $= 6 \times 3 \times 3$

$\mathbf{= 54\ cm^2}$

Tip: 3 × 3 × 3 = 27.

Task 2

PRACTICE · 9 MIN

SILVER *Solve*

Hint Cube: find the side first Prism: area × length

1. Cube volume 8 cm^3. Side?

2. Cube volume 1000 cm^3. Side?

3. Cube volume 64 cm^3. Surface area?

4. Cuboid 4 cm by 3 cm by 2 cm. Volume and surface area?

5. Cuboids 5 by 2 by 2 cm and 3 by 2 by 2 cm joined. Volume?

★ Cube SA 150 cm^2. Volume?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 2 cm
2. 10 cm
3. 96 cm^2
4. 24 cm^3 and 52 cm^2
5. 20 + 12 = 32 cm^3

★ 125 cm^3

Examples 5 and 6 — Multi-Step Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Write an equation SA: all six faces

Example 5

(4 marks)

A cuboid has volume 37.8 m^3.
It is 3.6 m long, 2.5 m wide.

Find its height, then its surface area.

Example 6

(2 marks)

A box 40 cm by 30 cm by 20 cm is filled with cubes of side 10 cm.

How many cubes fit?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Base: 3.6 × 2.5 = 9 m²

$9h = 37.8$

$h = 4.2$ m

$SA = 2(9 + 15.12 + 10.5)$

$\mathbf{= 69.24\ m^2}$

Tip: 3.6 × 4.2 = 15.12, 2.5 × 4.2 = 10.5.

Example 6

Count along each edge

cubes $= 4 \times 3 \times 2$

$\mathbf{= 24}$

Tip: 40 ÷ 10 = 4, 30 ÷ 10 = 3, 20 ÷ 10 = 2.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: multi-step problems*

Hint Write an equation SA: all six faces

1. Volume 60 cm^3, base 5 cm by 3 cm. Height?

2. Box 20 by 20 by 10 cm filled with 5 cm cubes. How many?

3. Tank 60 cm by 40 cm by 50 cm. Litres?

4. Cuboid 2 m by 1.5 m by 1 m. Surface area?

5. Paint that cuboid at £4 per m^2. Cost?

★ Tank 200 litres fills at 5 litres per minute. Time?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **4 cm**
2. **4 × 4 × 2 = 32**
3. **120 litres**
4. **13 m²**
5. **£52**

★ **40 minutes**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Convert 2500 ml to cm^3

SILVER Volume of a cuboid 3 cm by 4 cm by 5 cm

GOLD Convert 2000 cm^3 to litres

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **2500 cm^3**

SILVER **60 cm^3**

GOLD **2 litres**

Exam Practice — Capacity

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Same units first 1 m³ = 1000 litres

Question 1

(2 marks)

A jug holds 1.8 litres of water.

How many 300 ml glasses can be filled?

.....

Question 2

(3 marks)

A pool is 25 m by 10 m by 2 m.

How many litres does it hold?

..... litres

Exam Practice — Volume and Surface Area

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Cube: find the side first Prism: area × length

Question 3

(3 marks)

A cube has volume 125 cm^3.

Work out its surface area.

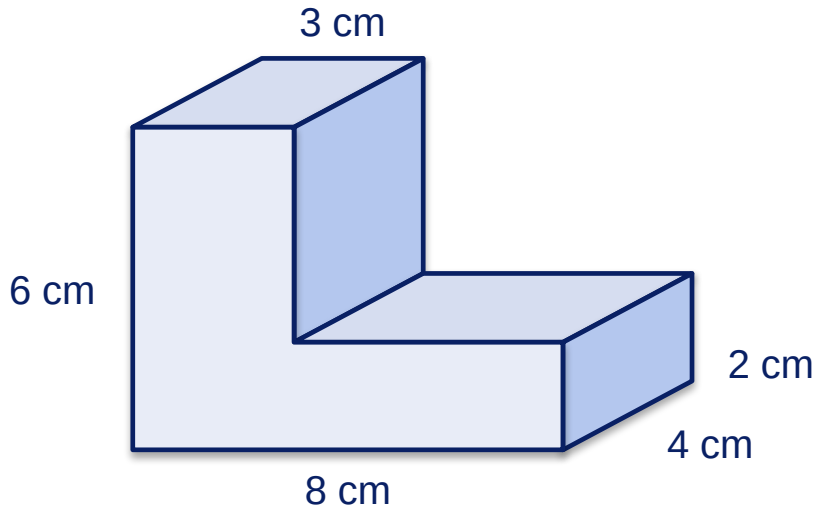
..... cm^2

Question 4

(3 marks)

Work out the volume of this prism.

Diagram NOT accurately drawn



..... cm^3

Exam Practice — Multi-Step Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Write an equation SA: all six faces

Question 5

(4 marks)

A cuboid has volume 48 m^3.
It is 4 m long and 3 m wide.

(a) Work out its height. (2)

..... m

(b) Work out its surface area. (2)

..... m^2

Question 6

(4 marks)

A tank is 60 cm by 40 cm by 30 cm.
Water flows in at 4 litres per minute.

How long does it take to fill the tank?

..... minutes

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

1.8 litres = 1800 ml — M1

glasses = 1800 ÷ 300 = 6 — A1

Tip: same units first.

Question 2

$V = 25 \times 10 \times 2$ — M1

$= 500\ m^3$ — A1

= 500 000 litres — A1

Tip: 1 m^3 = 1000 litres.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

side	= 5 cm	B1
SA	= 6 × 5 × 5	M1
	= 150 cm²	A1

Tip: 5 × 5 × 5 = 125.

Question 4

area	= 3 × 6 + 5 × 2 = 28 cm²	M1
V	= 28 × 4	M1
	= 112 cm³	A1

Tip: 8 − 3 = 5 cm.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $4 \times 3 \times h$	$= 48$	M1
$\boldsymbol{h}$	$\boldsymbol{= 4}$ **m**	A1
(b) SA	$= 2(12 + 16 + 12)$	M1
	$\boldsymbol{= 80}$ **m²**	A1

Tip: 4 × 4 = 16 and 3 × 4 = 12.

Question 6

V	$= 60 \times 40 \times 30$	M1
	$= 72\,000$ cm³ = 72 litres	A1
time	$= 72 \div 4$	M1
	= 18 minutes	A1

Tip: litres first, then divide by the rate.